

To find the Unit's Place Digit of a given Expression

When Number is in the form of Product

To find the unit digit in the product of two or more number we take unit digit of every number and then multiply them. Then, the unit digit of the resultant product is the unit digit of the product of original numbers.

When Number is in the form of Index

Let the exponential number of the form be a^n and $n \in I$.

1. In case, if a is any of (0, 1, 5, 6), then the unit's place digit is 0, 1, 5 and 6, respectively.
2. In case, if a is any of (4 and 9)
 - (i) and if power is odd, then the unit's place digit is 4 and 9, respectively.
 - (ii) and if power is even, then the unit's place digit is 6 and 1, respectively.
3. In case, if a is any of (2, 3, 7, 8), then see the following steps

Step I First, divide the exponent of a by 4.

Step II If any remainder comes on division. Put it as the power of a and get the result.

Step III If any remainder does not come on division. Put 4 as the power of a and get the result.

EXAMPLE 5. Find the unit digit of $207 \times 781 \times 39 \times 94$.

- a. 4 b. 2 c. 1 d. 5

Sol. b. Taking unit digit of every number and then multiplying them $= 7 \times 1 \times 9 \times 4 = 7 \times 36$

Again taking unit digit and then multiplying $= 7 \times 6 = 42$
 $\therefore$ Required unit digit = 2

EXAMPLE 6. What is the last digit in $7^{402} + 3^{402}$?

- a. 0 b. 4 c. 8 d. None of these

Sol. c. On division of 402 by 4, we get 2 as remainder.

$\therefore$ Last digit of $7^{402} = \text{Last digit of } 7^2 = \text{Last digit of } 49 = 9$
 Last digit of $3^{402} = \text{Last digit of } 3^2$
 $= \text{Last digit of } 9 = 9$

$\therefore$ Last digit of $7^{402} + 3^{402} = \text{Last digit of } (9 + 9) = 8$

DIVISION ON NUMBERS (DIVISION ALGORITHM)

Let ' a ' and ' b ' be two integers such that $b \neq 0$. On dividing ' a ' by ' b ', ' q ' will be the quotient and ' r ' will be the remainder, then the relationship between a , b , q and r is $a = bq + r$, where $0 \leq r < b$. Or in general, we have

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

EXAMPLE 7. When a positive integer n is divided by 5, the remainder is 2. What is the remainder when the number $3n$ is divided by 5?

- a. 1 b. 2 c. 3 d. 4

Sol. a. Let $n = 5q + 2$ and $3n = 3(5q + 2)$

$\Rightarrow 3n = 15q + 6 = 15q + (5 + 1) = 5(3q + 1) + 1$
 when $3n$ is divided by 5, then remainder is 1.

Divisibility Test

Divisor	Condition with Example
2	If the unit place of a number is '0' or divisible by 2 i.e. unit place is even. e.g. 17980, 314782, 6148, 316 etc.
3	If the sum of the digits of the given number is divisible by 3. e.g. 24375, here $2 + 4 + 3 + 7 + 5 = 21 \div 3 = 7$ Hence, 24375 is divisible by 3.
4	If last two digits of number is divisible by 4. e.g. 589372, 72 is divisible by 4. Hence, 589372 is also divisible by 4.
5	If digit at unit place is 5 or 0. e.g. 895, 700 etc.
6	If given number is divisibly by both 2 and 3. e.g. 759312. Here the last digit is divisible by 2. And $(7 + 5 + 9 + 3 + 1 + 2) = 27$, is divisible by 3. Hence, 759312 is divisible by 6.
7	If twice the number at units place is subtracted from rest of the digits and the remainder is divisible by 7. e.g. (a) $875 = 87 - 2(5) = 87 - 10 = 77 \div 7 = 11$ Hence, 875 is divisible by 7. (b) $5103 = 510 - 2(3) = 510 - 6 = 504 \div 7 = 72$ Hence, 5103 is divisible by 7. Note Trick is applicable for number greater than 99.
8	If last three digits are divisible by 8. e.g. (a) $96432 \rightarrow 432 \div 8 = 54$ (b) $16000 \rightarrow 000 \div 8 = 0$
9	If sum of all the digits are divisible by 9. e.g. $317349 \Rightarrow (3 + 1 + 7 + 3 + 4 + 9) = 27 \div 9 = 3$ Hence, 317349 is divisible by 9.
10	If last digit of a number is '0'. e.g. 130, 36980, etc.
11	If the difference between sum of digits at even places and sum of digits at odd places is divisible by 11. e.g. 10615. Sum of digits at odd place $= 1 + 6 + 5 = 12$ Sum of digits at even place $= 0 + 1 = 1$ Difference $= 12 - 1 = 11 \div 11 = 1$ So, 10615 is divisible by 11.

EXAMPLE 8. What is the remainder when 4^{1000} is divided by 7? 

- a. 1 b. 2 c. 4 d. 5

Sol. c. On division of $4^1, 4^2, 4^3, 4^4, 4^5, 4^6$, and 4^7 by 7,

we get remainders 4, 2, 1, 4, 2, 1 and 4 respectively.

Now, 4^4 gives us same remainder as 4^1 , so the cyclicity is of 3.

So, any power of 3 or a multiple of 3 will give a remainder of 1.

So, 4^{999} will give a remainder of 1.

$\therefore$ Remainder of $\frac{4^{1000}}{7} = 4$

Theorem of Divisibility

1. If N is a composite number of the form $N = a^p \cdot b^q \cdot c^r \dots$, where a, b and c are primes, then the number of divisors of N , represented by m is given by $m = (p+1)(q+1)(r+1)\dots$
2. The sum of the divisors of N , represented by S is given by

$$S = \frac{(a^{p+1} - 1)}{(a - 1)} \cdot \frac{(b^{q+1} - 1)}{(b - 1)} \cdot \frac{(c^{r+1} - 1)}{(c - 1)}$$

SOME IMPORTANT RESULTS ON DIVISION

1. If p divides q and r , then p also divides their sum and difference also.
2. For any natural number n , $(n^3 - n)$ is divisible by 6.
3. The product of three consecutive natural numbers is always divisible by 6.
4. $(x^m - a^m)$ is divisible by $(x + a)$ for even values of m .
5. $(x^m + a^m)$ is divisible by $(x + a)$ for odd values of m .
6. $(x^m - a^m)$ is divisible by $(x - a)$ for all values of m .

EXAMPLE 9. $19^5 + 21^5$ is divisible by

- a. Only 10 b. Only 30
c. Both 10 and 20 d. Neither 10 nor 20

Sol. c. We can check divisibility of $19^5 + 21^5$ by 10 by adding the unit digits of 9^5 and 1^5 which is equal to $9 + 1 = 10$. So, it must be divisible by 10.

Now, for divisibility by 20 we add 19 and 21 which is equal to 40.

So, it is clear that it is also divisible by 20.

So, $19^5 + 21^5$ is divisible by both 10 and 20.

Shortcut Methods for Multiplication

1. Multiplication of a Given Number by 9, 99 etc.

Method Place as many zeros at the right of the multiplicand as is the number of nines in the multiplier and subtract from the number, so formed from the multiplicand, to get the result. e.g.

- (i) Multiply 8886 by 9999

$$\text{So, } 8886 \times 9999 = 88860000 - 8886 = 88851114$$

- (ii) Multiply 56985 by 999

$$\text{So, } 56985 \times 999 = 56985000 - 56985 = 56928015$$

2. Multiplication of a Given Number by a Power of 5

Method Put as many zeros to the right of the multiplicand as is the power of 5 in the multiplier. Divide the number so formed by 2 raised to the same power as is the power of 5.

e.g. Multiply 6798 by 125

$$\begin{aligned} \text{Here, } 6798 \times 125 &= 6798 \times 5^3 = \frac{6798000}{2^3} \\ &= \frac{6798000}{8} = 849750 \end{aligned}$$

Important Identities

The identities given below are very useful for quick multiplication.

Here, if a, b and c are real numbers, then

1. (i) $(a + b)^2 = (a + b)(a + b) = a^2 + b^2 + 2ab$
(ii) $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
2. (i) $(a - b)^2 = (a - b)(a - b) = a^2 + b^2 - 2ab$
(ii) $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
3. (i) $(a + b) = \sqrt{(a - b)^2 + 4ab}$
(ii) $(a - b) = \sqrt{(a + b)^2 - 4ab}$
4. (i) $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
(ii) $(a + b)^2 - (a - b)^2 = 4ab$
5. $ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$
6. $a^2 - b^2 = (a - b)(a + b)$

EXAMPLE 10. The value of

$$\frac{1.073 \times 1.073 - 0.927 \times 0.927}{1.073 - 0.927} + \frac{(3^4)^4 \times 9^6}{(27)^7 \times 3^9} \text{ is}$$

- a. $2\frac{1}{3}$ b. $2\frac{1}{5}$
c. $2\frac{1}{9}$ d. 3

$$\begin{aligned} \text{Sol. c. } & \frac{1.073 \times 1.073 - 0.927 \times 0.927}{1.073 - 0.927} + \frac{(3^4)^4 \times (9)^6}{(27)^7 \times (3)^9} \\ &= \frac{(1.073)^2 - (0.927)^2}{1.073 - 0.927} + \frac{(3^4)^4 \times (3^2)^6}{(3^3)^7 \times (3)^9} \\ &= \frac{(1.073 + 0.927)(1.073 - 0.927)}{1.073 - 0.927} + \frac{3^{28}}{3^{30}} \\ &= 2 + \frac{1}{3^2} = 2 + \frac{1}{9} = 2\frac{1}{9} \end{aligned}$$

SIMPLIFICATION

The word simplification refers to a procedure of converting a complex arithmetical expression into a simple expression.

For simplifying an expression, follow the rule 'VBODMAS'. The letter of word means

V	—	Under line portion i.e. bar
B	—	Brackets
O	—	Orders or powers
D	—	Division
M	—	Multiplication
A	—	Addition
S	—	Subtraction

- The operations have to be carried out in the order, in which they appear in the word VBODMAS.
- The order of brackets to be simplified is $()$, $\{\}$, $[\]$.

EXAMPLE 11. Simply the following expression $52 - 4$ of $(17 - 12) + 4 \times 7$.

- a. 38 b. 85 c. 60 d. 72

Sol. c. $52 - 4$ of $(17 - 12) + 4 \times 7$
 $= 52 - 4$ of $5 + 4 \times 7$ (simplifying parenthesis)
 $= 52 - 4 \times 5 + 4 \times 7$ (simplifying of)
 $= 52 - 20 + 4 \times 7$ (simplifying multiplication)
 $= 52 - 20 + 28$ (simplifying multiplication)
 $= 32 + 28$ (simplifying subtraction)
 $= 60$ (simplifying addition)

PRACTICE EXERCISE

- Let 'a' and 'b' be natural number, not necessarily distinct. For all values of 'a' and 'b' the natural number would be
 (a) $(a + b)$ (b) a / b (c) $a - b$ (d) $\log(ab)$
- x and y are two natural numbers such that x is less than y , q is the quotient and r is the remainder when y is divided by x . Therefore,
 (a) $r = 0$ (b) $r < 0$ (c) $r > x$ (d) $0 \leq r < x$
- If p is a prime number and p divides ab i.e. p/ab , where 'a' and 'b' are integers, then
 (a) p/a or p/b (b) p/a and p/b
 (c) $p/a - b$ (d) None of these
- Which one of the following is a prime number?
 (a) 161 (b) 171
 (c) 173 (d) 221
- If n is a natural number, then $\sqrt{n}$ is
 (a) always a whole number
 (b) always a natural number
 (c) sometimes a natural number and sometimes an irrational number
 (d) always an irrational number
- The product of a rational number and an irrational number is
 (a) natural number (b) an irrational number
 (c) a composite number (d) a rational number
- What is the value of x for which x , $x + 1$, $x + 3$ are all prime numbers?
 (a) 0 (b) 1
 (c) 2 (d) 101
- In a division operation, the divisor is 5 times the quotient and twice the remainder. If the remainder is 15, then what is the dividend?
 (a) 175 (b) 185 (c) 195 (d) 250
- If $42 * 8$ is a multiple of 9, then the digit represented by $*$ is
 (a) 0 (b) 1 (c) 2 (d) 4
- The value at $*$ in $\frac{16}{7} \times \frac{16}{7} - \frac{*}{7} \times \frac{9}{7} + \frac{9}{7} \times \frac{9}{7} = 1$ is
 (a) 33 (b) -33 (c) -11 (d) 32
- The value of $10 \div 4 + 6 \times 4$ is
 (a) 4 (b) $1/4$ (c) 5 (d) None of these
- If $\frac{a}{4} = \frac{b}{5} = \frac{c}{6}$, then the value of $\frac{a + b + c}{b}$ is
 (a) 3 (b) 2 (c) 6 (d) 4
- The value of $[a - b + (b - a)] - [2a - 2b + (b - 2a)]$ is
 (a) $b - 2a$ (b) $a - 2b$ (c) $b + 2a$ (d) b
- If $3.325 \times 10^k = 0.0003325$, the value of 'k' is
 (a) 4 (b) -4 (c) -3 (d) -2
- If $\sqrt{[0.04 \times 0.4 \times x]} = 0.4 \times 0.04 \times \sqrt{y}$, then the value of x/y is
 (a) 0.0016 (b) 0.16 (c) 0.016 (d) 0.160
- The unit digit in the product $(127)^{170}$ is
 (a) 3 (b) 9 (c) 7 (d) 3
- The unit digit in the product $(7^{71} \times 6^{59} \times 3^{65})$ is
 (a) 6 (b) 2 (c) 4 (d) 1

- 18.** When n is divided by 4, the remainder is 3. What is the remainder when $2n$ is divided by 4?
 (a) 0 (b) 2 (c) 6 (d) 3
- 19.** If $\frac{\frac{1}{4} - \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} - \left(\frac{1}{6} - \frac{1}{48}\right)} \div \frac{\frac{1}{4} \times \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} \times \left(\frac{1}{6} - \frac{1}{48}\right)} = x$, the value of x is
 (a) $\frac{20}{21}$ (b) $-\frac{21}{20}$ (c) $\frac{21}{20}$ (d) $-\frac{20}{21}$
- 20.** If x is negative real number, then
 (a) $|x| = x$ (b) $|x| = -x$ (c) $|x| = \frac{1}{x}$ (d) $|x| = -\frac{1}{x}$
- 21.** If $\frac{x}{y} = \frac{3}{5}$, then the value of $\frac{x-y}{x+y}$ is
 (a) $-\frac{1}{4}$ (b) $\frac{1}{4}$ (c) $\frac{1}{6}$ (d) -6
- 22.** If x is real, then $\left|\frac{7-x}{3}\right| < 2$, if and only if
 (a) $1 < x < 13$ (b) $-1 < x < 13$ (c) $x < 13$ (d) $x > 13$
- 23.** A number is divisible by 25 only, if
 (a) the last digit of the number is zero
 (b) the last digit of the number is 5
 (c) the last two digit of the number is divisible by 15
 (d) the last two digit of the number is divisible by 25
- 24.** The value of $\frac{2.48 \times 2.48 - 1.52 \times 1.52}{0.96}$ is
 (a) 4 (b) 0.96 (c) 16 (d) 15.04
- 25.** Which one of the following is correct regarding the number 222222?
 (a) It is divisible by 3 but not divisible by 7
 (b) It is divisible by 3 and 7 but not divisible by 11
 (c) It is divisible by 2 and 7 but not divisible by 11
 (d) It is divisible by 3, 7 and 11
- 26.** What is the sum of positive integers less than 100 which leave a remainder 1 when divided by 3 and leave a remainder 2 when divided by 4?
 (a) 416 (b) 620 (c) 1250 (d) 1314
- 27.** What is the last digit in the expansion of $(2457)^{754}$?
 (a) 3 (b) 7 (c) 8 (d) 9
- 28.** A three-digit number is divisible by 11 and has its digit in the unit's place equal to 1. The number is 297 more than the number obtained by reversing the digits. What is the number?
 (a) 121 (b) 231 (c) 561 (d) 451
- 29.** The remainder on dividing given integers a and b by 7 are, respectively 5 and 4. What is the remainder when ab is divided by 7?
 (a) 3 (b) 4 (c) 5 (d) 6
- 30.** If k is any even positive integer, then $(k^2 + 2k)$ is
 (a) divisible by 24
 (b) divisible by 8 but may not be divisible by 24
 (c) divisible by 4 but may not be divisible by 8
 (d) divisible by 2 but may not be divisible by 4
- 31.** The number 2784936 is divisible by which one of the following numbers?
 (a) 86 (b) 87 (c) 88 (d) 89
- 32.** Which among the following is the largest four-digit number that is divisible by 88?
 (a) 9988 (b) 9966 (c) 9944 (d) 8888
- 33.** The pair of rational numbers that lies between $\frac{1}{4}$ and $\frac{3}{4}$ is
 (a) $\frac{262}{1000}, \frac{752}{1000}$ (b) $\frac{24}{100}, \frac{74}{100}$
 (c) $\frac{9}{40}, \frac{31}{40}$ (d) $\frac{252}{1000}, \frac{748}{1000}$
- 34.** By adding x to 1254934, the resulting number becomes divisible by 11, while adding y to 1254934 makes the resulting number divisible by 3. Which one of the following is the set of values for x and y ?
 (a) $x = 1, y = 1$ (b) $x = 1, y = -1$
 (c) $x = -1, y = 1$ (d) $x = -1, y = -1$
- 35.** What is the last digit in the expansion of 3^{4798} ?
 (a) 1 (b) 3 (c) 7 (d) 9
- 36.** What least value must be given to $\otimes$, so that the number $84705 \otimes 2$ is divisible by 9?
 (a) 0 (b) 1 (c) 2 (d) 3
- 37.** What is the total number of three digit numbers with unit digit 7 and divisible by 11?
 (a) 6 (b) 7 (c) 8 (d) 9
- 38.** Find the value of 'a' and 'b' $3\frac{7}{a} \times b\frac{3}{15} = 8$.
 (a) 2, 11 (b) 11, 2 (c) 1, 1 (d) 2, 1
- 39.** If a, b and c are real numbers such that $a < b$ and $c < 0$, then which of the statements is true?
 (a) $(a/c) < (b/c)$ (b) $ac < bc$
 (c) $(c/a) > (c/b)$ (d) $ac > bc$
- 40.** If we divide a positive integer by another positive integer, what is the resulting number?
 (a) It is always a natural number
 (b) It is always an integer
 (c) It is a rational number
 (d) It is an irrational number
- 41.** What can be said about the expansion of $2^{12n} - 6^{4n}$, where n is a positive integer?
 (a) Last digit is 4 (b) Last digit is 8
 (c) Last digit is 2 (d) Last two digits are zero

- 42.** If p is an integer, then every square integer is of the form
 (a) $2p$ or $(4p-1)$ (b) $4p$ or $(4p-1)$
 (c) $3p$ or $(3p+1)$ (d) $4p$ or $(4p+1)$
- 43.** If r and s are any real numbers such that $0 \leq s \leq 1$ and $r + s = 1$, then what is the maximum value of the product?
 (a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$
- 44.** A number when divided by 2, 3 or 5 gives remainder 1. The number is
 (a) 31 (b) 47 (c) 49 (d) 53
- 45.** The largest integer that divides product of any four consecutive integers is
 (a) 4 (b) 6 (c) 12 (d) 24
- 46.** Which one of the following three digit numbers divides 9238 and 7091 with the same remainder in each case?
 (a) 113 (b) 209 (c) 317 (d) 191
- 47.** For a positive integer n , define $d(n)$ = The number of positive divisors of n . What is the value of $d(d(d(12)))$?
 (a) 1 (b) 2 (c) 4 (d) None of these
- 48.** If three sides of a right angled triangle are integers in their lowest form, then one of its sides is always divisible by
 (a) 6 (b) 5 (c) 7 (d) None of these
- 49.** If $-1 \leq x \leq 3$ and $1 \leq y \leq 3$, then the maximum value of $(3y - 4x)$ is
 (a) 18 (b) 13 (c) 5 (d) -6
- 50.** If ' p ' is an integer greater than 3, then on dividing $p^{11} + 1$ by $p - 1$, we would get the remainder as
 (a) 2 (b) 0 (c) -2 (d) -1
- 51.** If A is the set of squares of natural numbers and x and y are any two element of A , then the correct statement is
 (a) $x + y$ belongs to A (b) $x - y$ belongs to A
 (c) $\frac{x}{y}$ belongs to A (d) xy belongs to A
- 52.** A ten-digit number is divisible by 4 as well as by 5. What could be the possible digit at the ten's place in the given number?
 (a) 0, 1, 2, 4, or 6 (b) 1, 2, 4, 6 or 8
 (c) 2, 3, 4, 6 or 8 (d) 0, 2, 4, 6 or 8
- 53.** If $x < 0 < y$, then which one of the following relations is correct?
 (a) $\frac{1}{x^2} < \frac{1}{xy} < \frac{1}{y^2}$ (b) $\frac{1}{x^2} > \frac{1}{xy} < \frac{1}{y^2}$
 (c) $\frac{1}{x} < \frac{1}{y}$ (d) $\frac{1}{x} > \frac{1}{y}$
- 54.** What will be the remainder when 19^{100} is divided by 20?
 (a) 19 (b) 20 (c) 3 (d) 1
- 55.** $7^{12} - 4^{12}$ is exactly divisible by which of the following number?
 (a) 34 (b) 33 (c) 36 (d) 35
- 56.** If $2^p + 3^q = 17$ and $2^{p+2} - 3^{q+1} = 5$, then find the value of p and q .
 (a) -2, 3 (b) 2, -3 (c) 3, 2 (d) 2, 3
- 57.** If $\frac{49}{15} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, where x, y and z are natural numbers, then what is z equal to?
 (a) 1 (b) 2 (c) 3
 (d) Cannot be determined due to insufficient data
- 58.** The least number which is a perfect square and has 540 as a factor is
 (a) 8100 (b) 6400 (c) 4900 (d) 3600
- 59.** If A is real and $1 + A + A^2 + A^3 = 40$, then A is equal to
 (a) -3 (b) -1 (c) 1 (d) 3
- 60.** How many factors of 1080 are perfect squares?
 (a) 4 (b) 6 (c) 8 (d) 5
- 61.** Consider the following statements:
 I. In a given whole number, if the sum of the odd numbered digit is equal to the sum of even numbered digits, then the number is divisible by 11.
 II. In a given whole number, if the difference of sum of odd numbered digits and even numbered digits is divisible by 11, then the number is divisible by 11.
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Both I and II
 (c) Only II (d) None of these
- 62.** Consider the following statements:
 A number $a_1a_2a_3a_4a_5$ is divisible by 9, if
 I. $a_1 + a_2 + a_3 + a_4 + a_5$ is divisible by 9.
 II. $a_1 - a_2 + a_3 - a_4 + a_5$ is divisible by 9.
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 63.** Consider the following statements:
 If p is a prime such that $p + 2$ is also a prime, then
 I. $p(p + 2) + 1$ is a perfect square.
 II. 12 is a divisor of $p + (p + 2)$, if $p > 3$